SOFTWARE REQUIREMENTS SPECIFICATION (SRS) & SYSTEM PROPOSAL

Diagnostic Centre Patient Management & Doctor Booking System

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Prepared by: Development Team
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1. INTRODUCTION

1.1 Purpose

This document defines the complete software requirements for a **Diagnostic Centre Patient Management and Doctor Booking System**—a web application integrating patient management, doctor appointment scheduling, payment processing, admin controls, and WhatsApp notification system.

1.2 Scope

The system will include:

- **Website** (Business/Marketing portal)
- **Patient Portal** (Self-service appointments, medical history)
- **Doctor Appointment Booking** (Search, book, reschedule, cancel)
- **Admin Dashboard** (User management, analytics, reporting)
- **WhatsApp Integration** (Appointment reminders, notifications)
- **Payment Gateway** (POS/API for consultation fees)

1.3 Intended Audience

- Patients
- Doctors/Medical Professionals
- Administrative Staff
- System Administrators
- Management/Stakeholders

1.4 Definitions & Acronyms

Term	Definition
SRS	Software Requirements Specification
MVP	Minimum Viable Product
HIPAA	Health Insurance Portability and Accountability Act
GDPR	General Data Protection Regulation
JWT	JSON Web Token
API	Application Programming Interface
POS	Point of Sale
EHR	Electronic Health Record
SMS	Short Message Service
OAuth2	Open Authorization 2.0 Protocol

2. OVERALL DESCRIPTION

2.1 Product Overview

A comprehensive web-based healthcare management platform enabling patients to search for doctors by specialty/location, book appointments, manage medical records, receive WhatsApp notifications, and pay for services. Doctors and admins gain tools to manage schedules, patient interactions, and business analytics.

2.2 Product Perspective

User Classes

User Type	Description	Key Capabilities
Patient	General public seeking healthcare	Register, search doctors, book appointments, view history, pay online
Doctor	Healthcare professional	Manage schedule, view appointments, update availability, access patient info
Admin	System administrator	User management, analytics, reporting, system configuration

User Type	Description	Key Capabilities
Clinic Staff	Administrative support	Manage schedules, patient check-in, billing

2.3 Operating Environment

Hardware Requirements

- **Server:** Cloud-based (AWS, Azure, GCP) with load balancing
- **Client:** Desktop, tablet, or mobile device with modern browser

Software Environment

- **Web Browsers:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- **Mobile:** iOS 12+ (Safari), Android 7+ (Chrome)
- **Server OS:** Linux (Ubuntu 20.04 LTS) or Windows Server 2019+

3. SPECIFIC FEATURES & FUNCTIONAL REQUIREMENTS

3.1 WEBSITE (Business/Marketing Portal)

3.1.1 Landing Page

- **FR 3.1.1.1:** Display clinic/diagnostic centre information (name, address, contact, hours)
- **FR 3.1.1.2:** Showcase list of doctors with specialties, qualifications, ratings
- **FR 3.1.1.3:** Display services offered with descriptions
- **FR 3.1.1.4:** Display testimonials and patient reviews
- **FR 3.1.1.5:** "Book Now" and "Learn More" call-to-action buttons
- **FR 3.1.1.6:** Mobile-responsive design

3.1.2 About Us Section

- **FR 3.1.2.1:** Clinic history, mission, and values
- **FR 3.1.2.2:** Meet the team page with doctor profiles
- **FR 3.1.2.3:** Certifications and accreditations

3.1.3 Services Page

- **FR 3.1.3.1:** List all diagnostic services/specialties
- **FR 3.1.3.2:** Detailed descriptions with pricing
- **FR 3.1.3.3:** Service-specific FAQs

3.1.4 Contact Page

- **FR 3.1.4.1:** Contact form (Name, Email, Phone, Message)
- **FR 3.1.4.2:** Clinic location map (Google Maps integration)

- **FR 3.1.4.3:** Multiple contact options (phone, email, WhatsApp)

3.2 PATIENT REGISTRATION & AUTHENTICATION

3.2.1 User Registration

- **FR 3.2.1.1:** Patients can register with Email/Phone number
- **FR 3.2.1.2:** Email verification via OTP
- **FR 3.2.1.3:** Phone verification via SMS OTP
- **FR 3.2.1.4:** Password strength validation (min 8 chars, mixed case, numbers, special chars)
- **FR 3.2.1.5:** Terms & conditions acceptance checkbox
- **FR 3.2.1.6:** Option to sign up via Google/Facebook (OAuth2)

3.2.2 Login

- **FR 3.2.2.1:** Email/Phone + Password login
- **FR 3.2.2.2:** "Remember me" functionality (14-day session)
- **FR 3.2.2.3:** Social login (Google, Facebook)
- **FR 3.2.2.4:** Forgot password with email/SMS reset link

3.2.3 Multi-Factor Authentication (Optional for MVP+)

- **FR 3.2.3.1:** Optional 2FA via SMS OTP
- **FR 3.2.3.2:** Optional 2FA via email OTP

3.3 PATIENT PROFILE MANAGEMENT

3.3.1 Profile Information

- **FR 3.3.1.1:** Display personal info (Name, DOB, Gender, Phone, Email)
- **FR 3.3.1.2:** Edit profile details
- **FR 3.3.1.3:** Upload profile picture
- **FR 3.3.1.4:** Address management (home, work, clinic)
- **FR 3.3.1.5:** Emergency contact details

3.3.2 Medical History

- **FR 3.3.2.1:** Record chronic conditions (diabetes, hypertension, etc.)
- **FR 3.3.2.2:** List allergies and medication intolerances
- **FR 3.3.2.3:** Current medications list
- **FR 3.3.2.4:** Previous medical reports/documents upload
- **FR 3.3.2.5:** Insurance information (if applicable)

3.3.3 Preferences

- **FR 3.3.3.1:** Language preference (English, Hindi, etc.)
- **FR 3.3.3.2:** Notification preferences (Email, SMS, WhatsApp, Push)
- **FR 3.3.3.3:** Appointment reminder timing (24h, 2h before)

3.4 DOCTOR SEARCH & DISCOVERY

3.4.1 Doctor Listing

- **FR 3.4.1.1:** Display all doctors with profile cards (name, specialty, qualification, rating, review count)
- **FR 3.4.1.2:** Pagination (12-20 doctors per page)
- **FR 3.4.1.3:** Doctor availability status indicator (Available/On Leave/Busy)

3.4.2 Search & Filtering

- **FR 3.4.2.1:** Search by doctor name
- **FR 3.4.2.2:** Filter by specialty (Cardiology, Orthopedics, etc.)
- **FR 3.4.2.3:** Filter by availability (Today, Tomorrow, This Week, This Month)
- **FR 3.4.2.4:** Filter by consultation type (In-person, Telemedicine, Home visit)
- **FR 3.4.2.5:** Sort by rating, experience, or newest
- **FR 3.4.2.6:** Search by location/clinic branch

3.4.3 Doctor Profile Page

- **FR 3.4.3.1:** Display full doctor information (bio, qualifications, specializations)
- **FR 3.4.3.2:** Professional certifications and licenses
- **FR 3.4.3.3:** Experience summary
- **FR 3.4.3.4:** Patient reviews and ratings (with review text and date)
- **FR 3.4.3.5:** Available time slots for next 30 days
- **FR 3.4.3.6:** Consultation fee display
- **FR 3.4.3.7:** "Book Appointment" button

3.5 APPOINTMENT BOOKING & MANAGEMENT

3.5.1 Appointment Booking Flow

- **FR 3.5.1.1:** Patient selects doctor → chooses date → selects time slot
- **FR 3.5.1.2:** System shows available slots in 15/30-min intervals
- **FR 3.5.1.3:** Display appointment fee before confirmation
- **FR 3.5.1.4:** Allow notes/complaints section (optional)
- **FR 3.5.1.5:** Display confirmation with booking reference ID
- **FR 3.5.1.6:** Prevent double-booking (transactional database ensures uniqueness)

3.5.2 Appointment Management (View/Reschedule/Cancel)

- **FR 3.5.2.1:** Patient can view all upcoming appointments
- **FR 3.5.2.2:** View appointment details (doctor, date, time, location, fee)
- **FR 3.5.2.3:** Reschedule appointment (select new date/time; 24-hour notice required)
- **FR 3.5.2.4:** Cancel appointment (24-hour notice required; refund policy applies)
- **FR 3.5.2.5:** View appointment history (past appointments with outcomes)
- **FR 3.5.2.6:** Automated conflict detection and prevention

3.5.3 Appointment Status Tracking

- **FR 3.5.3.1:** Status flow: Booked → Confirmed → Completed → Cancelled
- **FR 3.5.3.2:** Real-time status updates visible to patient

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3.6 APPOINTMENT REMINDERS & NOTIFICATIONS

3.6.1 WhatsApp Integration

- **FR 3.6.1.1:** Send WhatsApp reminder 24 hours before appointment
- **FR 3.6.1.2:** Send WhatsApp reminder 2 hours before appointment
- **FR 3.6.1.3:** Use pre-approved WhatsApp message templates
- **FR 3.6.1.4:** Include appointment details (doctor name, date, time, location)
- **FR 3.6.1.5:** Include reschedule/cancel links in message
- **FR 3.6.1.6:** Track message delivery status

3.6.2 Email Notifications

- **FR 3.6.2.1:** Send confirmation email on booking
- **FR 3.6.2.2:** Send reminder email 24 hours before
- **FR 3.6.2.3:** Send cancellation confirmation
- **FR 3.6.2.4:** Send rescheduling confirmation

3.6.3 SMS Notifications (Optional)

- **FR 3.6.3.1:** Send SMS reminder for patients without WhatsApp

- **FR 3.6.3.2:** One-time password (OTP) for authentication

3.6.4 Notification Preferences

- **FR 3.6.4.1:** Patient can enable/disable each notification type
- **FR 3.6.4.2:** Patient can set preferred reminder times

3.7 PAYMENT & BILLING

3.7.1 Payment Gateway Integration

- **FR 3.7.1.1:** Support multiple payment methods (Credit/Debit Card, Digital Wallets)
- **FR 3.7.1.2:** Integrate Stripe, Razorpay, or PayPal
- **FR 3.7.1.3:** PCI-DSS compliance for secure payment processing
- **FR 3.7.1.4:** Encrypted payment data transmission

3.7.2 Consultation Fees

- **FR 3.7.2.1:** Display consultation fee at booking stage
- **FR 3.7.2.2:** Allow setting variable fees per doctor/specialty
- **FR 3.7.2.3:** Support discounts and promotional codes (admin-managed)

3.7.3 Payment Processing

- **FR 3.7.3.1:** Process payment at booking or before appointment (configurable)

- **FR 3.7.3.2:** Payment confirmation email with invoice
- **FR 3.7.3.3:** Failed payment retry mechanism (3 attempts)
- **FR 3.7.3.4:** Payment status: Pending → Completed/Failed

3.7.4 Refunds & Cancellation Policy

- **FR 3.7.4.1:** Full refund if cancelled >24 hours before
- **FR 3.7.4.2:** 50% refund if cancelled 6-24 hours before
- **FR 3.7.4.3:** No refund if cancelled <6 hours before
- **FR 3.7.4.4:** Automatic refund processing on cancellation
- **FR 3.7.4.5:** Refund status tracking

3.7.5 Invoicing & Receipt

- **FR 3.7.5.1:** Generate invoice with itemized charges
- **FR 3.7.5.2:** Download invoice as PDF
- **FR 3.7.5.3:** Email invoice automatically
- **FR 3.7.5.4:** Support multiple currencies (optional)

3.8 RATINGS & REVIEWS

3.8.1 Post-Appointment Reviews

- **FR 3.8.1.1:** Send automated request for review 24 hours after appointment
- **FR 3.8.1.2:** Star rating (1-5 stars with visual feedback)
- **FR 3.8.1.3:** Text review (optional, 10-500 characters)

- **FR 3.8.1.4:** Verified badge for actual appointment attendees

3.8.2 Review Management

- **FR 3.8.2.1:** Display reviews on doctor profile
- **FR 3.8.2.2:** Average rating calculation
- **FR 3.8.2.3:** Sort reviews (Newest, Highest, Lowest)
- **FR 3.8.2.4:** Doctor response to reviews (optional)

3.9 DOCTOR PORTAL

3.9.1 Doctor Registration & Authentication

- **FR 3.9.1.1:** Registration with email and license verification (admin approval required)
- **FR 3.9.1.2:** Secure login (Email + Password, OAuth2)
- **FR 3.9.1.3:** Password reset functionality

3.9.2 Doctor Profile Management

- **FR 3.9.2.1:** Manage professional information (qualifications, experience, bio)
- **FR 3.9.2.2:** Upload/update profile photo and documents
- **FR 3.9.2.3:** Set consultation fees
- **FR 3.9.2.4:** Manage specialties and services offered

3.9.3 Schedule Management

- **FR 3.9.3.1:** Set working hours (days and time slots)
- **FR 3.9.3.2:** Mark available/unavailable slots
- **FR 3.9.3.3:** Set break times
- **FR 3.9.3.4:** Mark vacation/leave periods (recurring or one-time)
- **FR 3.9.3.5:** Import schedule from calendar (optional)

3.9.4 Appointment Management

- **FR 3.9.4.1:** View all appointments (upcoming, past, cancelled)
- **FR 3.9.4.2:** View appointment details with patient info and complaints
- **FR 3.9.4.3:** Mark appointment as completed
- **FR 3.9.4.4:** Add notes/observations post-appointment
- **FR 3.9.4.5:** Reschedule or cancel appointments
- **FR 3.9.4.6:** Send custom messages to patients

3.9.5 Patient Interaction

- **FR 3.9.5.1:** View patient medical history
- **FR 3.9.5.2:** Add prescription/medical notes
- **FR 3.9.5.3:** Request additional tests or follow-ups
- **FR 3.9.5.4:** Send secure messages to patients (in-app, email, WhatsApp)

3.9.6 Analytics & Reports

- **FR 3.9.6.1:** View appointment statistics (booked, completed, cancelled, no-show)

- **FR 3.9.6.2:** Revenue/earnings summary (commission-based or consultation-fee based)
- **FR 3.9.6.3:** Patient satisfaction ratings
- **FR 3.9.6.4:** Download reports (monthly, quarterly, annual)

3.10 ADMIN DASHBOARD

3.10.1 User Management

- **FR 3.10.1.1:** View all patients (with filters: active, inactive, registration date)
- **FR 3.10.1.2:** View all doctors (with verification status)
- **FR 3.10.1.3:** Verify/approve doctor registrations
- **FR 3.10.1.4:** Disable/suspend user accounts
- **FR 3.10.1.5:** View and manage clinic staff accounts

3.10.2 Appointment Management

- **FR 3.10.2.1:** View all appointments (global, with filters by doctor, date, status)
- **FR 3.10.2.2:** Create/edit/delete appointments (admin override)
- **FR 3.10.2.3:** Manage no-shows and cancellations
- **FR 3.10.2.4:** Bulk appointment operations (import from spreadsheet)

3.10.3 Financial Management

- **FR 3.10.3.1:** View total revenue and earnings

- **FR 3.10.3.2:** Doctor commission management (percentage or fixed rate)
- **FR 3.10.3.3:** Payment reconciliation and settlement
- **FR 3.10.3.4:** Refund request management
- **FR 3.10.3.5:** Transaction history with filtering (date, doctor, status, type)

3.10.4 Analytics & Reporting

- **FR 3.10.4.1:** Dashboard with KPIs (total appointments, revenue, active doctors, active patients)
- **FR 3.10.4.2:** Appointment trends (daily, weekly, monthly charts)
- **FR 3.10.4.3:** Doctor performance metrics (avg rating, appointment count, no-show rate)
- **FR 3.10.4.4:** Patient acquisition and retention metrics
- **FR 3.10.4.5:** Export reports (PDF, Excel, CSV)

3.10.5 System Configuration

- **FR 3.10.5.1:** Manage consultation fee structure
- **FR 3.10.5.2:** Set/update refund policies
- **FR 3.10.5.3:** Configure notification settings
- **FR 3.10.5.4:** Manage promotional codes and discounts
- **FR 3.10.5.5:** Set cancellation and rescheduling deadlines
- **FR 3.10.5.6:** Configure payment gateway settings

3.10.6 Content Management

- **FR 3.10.6.1:** Manage website content (About Us, Services, FAQ)

- **FR 3.10.6.2:** Manage doctor profiles (bulk upload, edit, delete)
- **FR 3.10.6.3:** Manage clinic branches and locations
- **FR 3.10.6.4:** Upload testimonials and promotional banners

3.10.7 Audit & Logs

- **FR 3.10.7.1:** View system activity logs
- **FR 3.10.7.2:** Admin action tracking (who did what, when)
- **FR 3.10.7.3:** Login history of all users

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4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

4.1.1 Website (Public-Facing)

- **Landing Page:** Hero section, featured doctors, services, testimonials, CTA buttons
- **Doctor Listing:** Grid/List view with filters and search
- **Appointment Booking Wizard:** Multi-step form (doctor → date → time → confirm)
- **Contact Page:** Contact form and map

4.1.2 Patient Portal

- **Login Page:** Email/Phone + Password with social login options
- **Dashboard:** Upcoming appointments, quick actions (Book, Reschedule, View History)
- **Profile:** Editable personal and medical information
- **Appointments:** List with filters, detailed view, reschedule/cancel options
- **Payment History:** Invoices and transaction details
- **Notifications:** Inbox for appointment reminders and updates

4.1.3 Doctor Portal

- **Login Page:** Email + Password
- **Dashboard:** Today's appointments, quick stats
- **Schedule:** Calendar view with available/unavailable slots
- **Appointments:** List view with patient details and notes
- **Patients:** Patient list with medical history
- **Analytics:** Basic performance metrics
- **Profile:** Professional information and settings

4.1.4 Admin Dashboard

- **Dashboard:** KPIs, charts, recent activities
- **Users:** Doctor and patient management tables
- **Appointments:** Global appointment list with filters
- **Finance:** Revenue charts, payment reconciliation
- **Reports:** Analytics and export options
- **Settings:** Configuration options

4.2 Hardware Interfaces

- **Mobile Devices:** Touch-friendly UI with responsive design (breakpoints at 320px, 768px, 1024px)
- **Desktop:** Full-featured interface for 1920×1080 and above

4.3 Software Interfaces

4.3.1 Third-Party APIs

Service	Purpose	Endpoint Type
Stripe/Razorpay/PayPal	Payment processing	REST API
Twilio/MessageBird	WhatsApp & SMS	REST API
SendGrid/AWS SES	Email delivery	REST API
Google Maps	Location services	REST API/JavaScript SDK
Firebase	Push notifications (optional)	REST API/SDK
Google OAuth2	Social authentication	OAuth2
Facebook Login	Social authentication	OAuth2

4.3.2 Internal APIs

All internal APIs will follow **RESTful conventions** with JSON payloads:

Base URL: `https://api.diagnostic-centre.com/v1`

Authentication

Code

```
POST /auth/register
POST /auth/login
POST /auth/logout
POST /auth/refresh-token
POST /auth/forgot-password
```

Patient APIs

Code

```
GET /patients/{id}
PUT /patients/{id}
GET /patients/{id}/appointments
POST /patients/{id}/appointments
PUT /appointments/{appointmentId}
DELETE /appointments/{appointmentId}
POST /appointments/{appointmentId}/review
GET /appointments/{appointmentId}/invoice
```

Doctor APIs

```
GET /doctors
GET /doctors/{id}
GET /doctors/{id}/schedule
PUT /doctors/{id}/schedule
GET /doctors/{id}/appointments
PUT /doctors/{id}/appointments/{appointmentId}
```

Admin APIs

```
GET /admin/dashboard/stats
GET /admin/users
PUT /admin/users/{id}
GET /admin/appointments
POST /admin/appointments
GET /admin/analytics/revenue
```

4.4 Communication Interfaces

Protocol	Usage
HTTPS/TLS 1.2+	All API communication (encrypted)

Protocol	Usage
WebSocket	Real-time notifications (optional, for MVP+)
REST	Standard API communication
JWT	Stateless authentication tokens

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5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance

- **Page Load Time:** ≤ 3 seconds on 4G networks
- **API Response Time:** ≤ 500ms for 95th percentile
- **Database Query Time:** ≤ 100ms for standard queries
- **Concurrent Users:** Support 5,000 concurrent users (MVP)
- **Peak Load:** Support 50,000 concurrent users (with scaling)
- **Image Optimization:** Compressed and lazy-loaded images

5.2 Security

Data Protection

- **Encryption in Transit:** TLS 1.2+ for all connections

- **Encryption at Rest:** AES-256 for sensitive data (passwords, medical records)
- **Password Hashing:** bcrypt with salt and cost factor ≥ 12
- **Sensitive Fields:** Credit card data tokenized (not stored locally)

Authentication & Authorization

- **JWT Tokens:** 15-minute expiry for access tokens, 7-day for refresh tokens
- **OAuth2:** Support Google and Facebook login
- **RBAC:** Role-Based Access Control (Patient, Doctor, Admin, Staff)
- **MFA:** Optional 2FA via SMS/Email

HIPAA & GDPR Compliance

- **Medical Records:** Encrypted storage and transmission
- **Data Retention:** Comply with local regulations (e.g., 6-7 years for medical records)
- **Data Deletion:** GDPR "Right to be Forgotten" implementation
- **Audit Logs:** All access to sensitive data logged
- **Consent Management:** Explicit consent for data processing

API Security

- **Rate Limiting:** 100 requests per minute per IP/user
- **Input Validation:** Whitelist-based, prevent SQL injection and XSS
- **CORS:** Configured for allowed origins only
- **API Keys:** Rotate quarterly

5.3 Reliability

Uptime & Availability

- **Target Uptime:** 99.5% (SLA)
- **Backup & Recovery:** Daily automated backups, 4-hour RPO (Recovery Point Objective)
- **Disaster Recovery:** Hot standby or multi-region redundancy
- **Load Balancing:** Distribute traffic across multiple servers

Error Handling

- **Graceful Degradation:** System remains partially functional during partial outages
- **Circuit Breaker:** Third-party API failures don't crash the system
- **Retry Logic:** Auto-retry failed API calls (exponential backoff)
- **Error Logging:** All errors logged with context and severity

5.4 Scalability

- **Horizontal Scaling:** Stateless backend allows easy scaling
- **Database:** Optimize queries, use indexing; consider read replicas for reporting
- **Caching:** Redis for session storage and frequently accessed data
- **CDN:** Serve static assets from CDN (images, CSS, JS)

5.5 Usability

- **Accessibility:** WCAG 2.1 AA compliance (color contrast, keyboard navigation)
- **Mobile-First:** Responsive design, touch-friendly buttons (min 44×44 px)
- **Internationalization:** Support multiple languages (English, Hindi, Spanish)
- **Offline Fallback:** Graceful degradation for offline scenarios

5.6 Maintainability

- **Code Documentation:** JSDoc/Swagger for APIs, inline comments for complex logic
- **Testing:** ≥80% code coverage (Unit, Integration, E2E tests)
- **DevOps:** CI/CD pipeline (GitHub Actions, Jenkins)
- **Monitoring:** Logs (ELK Stack), metrics (Prometheus), traces (Jaeger)

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6. SYSTEM ARCHITECTURE & TECH STACK

6.1 Architecture Overview

○ Code

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│                        CLIENT LAYER                              │
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│  │  Web App            │  │  Mobile App         │  │  Admin
│  │  (React/Vue)        │  │  (React Native)     │  │  Portal
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│                    APPLICATION LAYER                             │
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│  │ Microservices (or Modular Monolith)
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│   │
│   │ ├─ User Service (Auth, Profile)
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│   │ ├─ Appointment Service
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│   │ ├─ Doctor Service
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STORAGE    │
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6.2 Recommended Tech Stack

Frontend (Client-Side)

Layer	Technology	Reason
Framework	React 18+ or Vue 3	Large ecosystem, component reusability
State Management	Redux Toolkit / Pinia	Predictable state, middleware support

Layer	Technology	Reason
HTTP Client	Axios / Fetch API	Simple API calls with interceptors
Styling	Tailwind CSS / Material-UI	Responsive, pre-built components
Form Handling	React Hook Form	Lightweight, minimal re-renders
Routing	React Router v6	Built-in SPA navigation
Build Tool	Vite	Fast, modern bundler
Testing	Jest + React Testing Library	Unit and integration testing
Linting	ESLint + Prettier	Code quality and formatting
Mobile	React Native / Flutter	Cross-platform iOS/Android app

Backend (Server-Side)

Layer	Technology	Reason
Runtime	Node.js 18+ LTS	Non-blocking I/O, fast development
Framework	Express.js or NestJS	Lightweight, modular, scalable

Layer	Technology	Reason
Language	TypeScript	Type safety, better maintainability
Database	PostgreSQL 14+	ACID compliance, JSON support, reliability
ORM	Sequelize / TypeORM / Prisma	Abstract DB layer, migrations, type safety
Caching	Redis 6+	Session storage, rate limiting, caching
Message Queue	RabbitMQ / AWS SQS	Async processing (notifications, emails)
Authentication	Passport.js / JWT	OAuth2, JWT tokens, multi-strategy
Logging	Winston / Pino	Structured logs with levels
Validation	Joi / Zod	Input validation, schema enforcement
Testing	Jest + Supertest	Unit, integration, API endpoint testing
API Documentation	Swagger/OpenAPI	Interactive API docs (Swagger UI)
Deployment	Docker + Kubernetes	Containerized, orchestrated

Layer	Technology	Reason
		deployment

DevOps & Infrastructure

Component	Technology	Reason
Cloud Provider	AWS / Google Cloud / Azure	Managed services, scalability, global reach
Containerization	Docker	Consistent environments, easy deployment
Orchestration	Kubernetes / ECS	Auto-scaling, load balancing, resilience
CI/CD	GitHub Actions / Jenkins	Automated testing, build, deployment
Monitoring	Prometheus + Grafana	Metrics collection, dashboards
Logging	ELK Stack (Elasticsearch, Logstash, Kibana)	Centralized logging, analysis
APM	DataDog / New Relic	Performance monitoring, error tracking

Component	Technology	Reason
Secrets Management	Vault / AWS Secrets Manager	Secure credential storage

Payment & Notification Services (Third-Party)

Service	Technology	Purpose
Payment	Stripe / Razorpay API	Process credit/debit cards, digital wallets
WhatsApp	Twilio / MessageBird API	Send appointment reminders, notifications
Email	SendGrid / AWS SES	Transactional emails
SMS	Twilio / AWS SNS	OTP, SMS notifications

6.3 Proposed Technology Choices (Recommended)

Frontend:

Code

```
React 18 + TypeScript + Vite + Redux Toolkit + Axios +
```

```
Tailwind CSS + React Router
```

Backend:

Code

```
Node.js + Express.js + TypeScript + PostgreSQL + Prisma
ORM + Redis + RabbitMQ
```

Mobile:

Code

```
React Native (for code sharing between iOS/Android) or
Flutter (for better performance)
```

DevOps:

Code

```
Docker + Kubernetes (or AWS ECS) + GitHub Actions +
Prometheus + ELK Stack
```

---

7. DATABASE DESIGN

7.1 Database Technology

- **Primary Database:** PostgreSQL 14+ (ACID compliance, JSON support)
- **Caching Layer:** Redis 6+ (Session, Rate Limit data)
- **Search (Optional):** Elasticsearch (Full-text search on doctor reviews)

**7.2 Core Database Schema

Users Table

SQL

```sql
CREATE TABLE users (
  id UUID PRIMARY KEY,
  email VARCHAR(255) UNIQUE NOT NULL,
  phone_number VARCHAR(20) UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  first_name VARCHAR(100) NOT NULL,
  last_name VARCHAR(100) NOT NULL,
  date_of_birth DATE,
  gender ENUM('M', 'F', 'O'),
  profile_photo_url VARCHAR(500),
  user_type ENUM('patient', 'doctor', 'admin', 'staff')
NOT NULL,
  status ENUM('active', 'inactive', 'suspended') DEFAULT
'active',
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW(),
```

```sql
  deleted_at TIMESTAMP
);
```

Patients Table

SQL

```sql
CREATE TABLE patients (
  id UUID PRIMARY KEY,
  user_id UUID NOT NULL REFERENCES users(id),
  blood_group VARCHAR(10),
  allergies TEXT,
  chronic_conditions TEXT,
  current_medications JSONB,
  emergency_contact_name VARCHAR(100),
  emergency_contact_phone VARCHAR(20),
  insurance_provider VARCHAR(100),
  insurance_policy_number VARCHAR(50),
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

Doctors Table

SQL

```sql
CREATE TABLE doctors (
  id UUID PRIMARY KEY,
```

```sql
  user_id UUID NOT NULL REFERENCES users(id),
  license_number VARCHAR(100) UNIQUE NOT NULL,
  license_expiry DATE NOT NULL,
  specialization VARCHAR(100) NOT NULL,
  qualification JSONB,
  experience_years INT,
  clinic_location VARCHAR(255),
  consultation_fee DECIMAL(10, 2),
  rating DECIMAL(3, 2) DEFAULT 0,
  total_reviews INT DEFAULT 0,
  is_verified BOOLEAN DEFAULT FALSE,
  verification_status ENUM('pending', 'approved',
'rejected') DEFAULT 'pending',
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

Appointments Table

SQL

```sql
CREATE TABLE appointments (
  id UUID PRIMARY KEY,
  patient_id UUID NOT NULL REFERENCES patients(id),
  doctor_id UUID NOT NULL REFERENCES doctors(id),
  appointment_date DATE NOT NULL,
  appointment_time TIME NOT NULL,
  appointment_type ENUM('in-person', 'telemedicine',
'home-visit'),
  status ENUM('booked', 'confirmed', 'completed',
```

```sql
'cancelled', 'no-show') DEFAULT 'booked',
  complaints TEXT,
  notes TEXT,
  fee DECIMAL(10, 2),
  payment_status ENUM('pending', 'completed',
'refunded') DEFAULT 'pending',
  cancellation_reason VARCHAR(255),
  cancelled_by ENUM('patient', 'doctor', 'admin'),
  cancelled_at TIMESTAMP,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW(),
  UNIQUE(doctor_id, appointment_date, appointment_time)
);
```

Reviews Table

SQL

```sql
CREATE TABLE reviews (
  id UUID PRIMARY KEY,
  appointment_id UUID NOT NULL REFERENCES
appointments(id),
  patient_id UUID NOT NULL REFERENCES patients(id),
  doctor_id UUID NOT NULL REFERENCES doctors(id),
  rating INT NOT NULL CHECK (rating >= 1 AND rating <=
5),
  review_text TEXT,
  is_verified BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT NOW(),
```

```
  updated_at TIMESTAMP DEFAULT NOW()
);
```

Payments Table

SQL

```sql
CREATE TABLE payments (
  id UUID PRIMARY KEY,
  appointment_id UUID NOT NULL REFERENCES
appointments(id),
  patient_id UUID NOT NULL REFERENCES patients(id),
  amount DECIMAL(10, 2) NOT NULL,
  currency VARCHAR(3) DEFAULT 'USD',
  payment_method ENUM('credit_card', 'debit_card',
'wallet', 'upi'),
  payment_gateway ENUM('stripe', 'razorpay', 'paypal'),
  gateway_transaction_id VARCHAR(255),
  status ENUM('pending', 'completed', 'failed',
'refunded') DEFAULT 'pending',
  refund_amount DECIMAL(10, 2),
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

Doctor Schedules Table

SQL

```sql
CREATE TABLE doctor_schedules (
  id UUID PRIMARY KEY,
  doctor_id UUID NOT NULL REFERENCES doctors(id),
  day_of_week INT (1-7, Monday=1),
  start_time TIME NOT NULL,
  end_time TIME NOT NULL,
  slot_duration_minutes INT DEFAULT 30,
  is_available BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

Notifications Table

SQL

```sql
CREATE TABLE notifications (
  id UUID PRIMARY KEY,
  user_id UUID NOT NULL REFERENCES users(id),
  type ENUM('appointment_reminder',
'appointment_confirmed', 'booking_confirmation',
'payment_receipt'),
  channel ENUM('email', 'sms', 'whatsapp', 'push'),
  content TEXT NOT NULL,
  sent_at TIMESTAMP,
  delivery_status ENUM('pending', 'sent', 'failed')
DEFAULT 'pending',
  created_at TIMESTAMP DEFAULT NOW()
);
```

Audit Logs Table

SQL

```sql
CREATE TABLE audit_logs (
  id UUID PRIMARY KEY,
  user_id UUID REFERENCES users(id),
  action VARCHAR(100) NOT NULL,
  entity_type VARCHAR(50),
  entity_id UUID,
  old_value JSONB,
  new_value JSONB,
  ip_address VARCHAR(45),
  created_at TIMESTAMP DEFAULT NOW()
);
```

---

8. DEVELOPMENT ROADMAP & TIMELINE

8.1 Project Phases

Phase 1: MVP (Months 1-3)

Goal: Core functionality for patient booking and doctor management

Deliverables:

- User authentication (patient & doctor registration/login)
- Doctor listing and search
- Appointment booking
- Basic email notifications
- Admin dashboard (basic user and appointment management)
- Payment gateway integration (Stripe/Razorpay)
- Responsive website

Team Size: 2 Backend devs, 2 Frontend devs, 1 QA, 1 DevOps

Key Technologies: React, Node.js + Express, PostgreSQL

Phase 2: Core Features (Months 4-6)

Goal: Add WhatsApp integration, reviews, and enhanced functionality

Deliverables:

- WhatsApp notification integration (Twilio)
- Patient medical history and profile management
- Ratings & reviews system
- Doctor schedule management
- Appointment rescheduling and cancellation with refunds
- Enhanced admin analytics
- Email notifications (SendGrid)

Team Addition: +1 Backend dev (total 3)

Phase 3: Enhancement & Scaling (Months 7-9)

Goal: Telemedicine, advanced features, and performance optimization

Deliverables:

- Telemedicine support (video call integration with Zoom API)
- Mobile app (React Native or Flutter)
- Doctor appointment analytics
- SMS OTP authentication
- Advanced search and filtering
- Performance optimization (caching, CDN)
- Load testing and optimization

Team Addition: +1 Mobile dev (total 6)

Phase 4: Production & Maintenance (Month 10+)

Goal: Launch, monitor, and maintain

Activities:

- UAT and bug fixes
- Security audit and penetration testing
- Production deployment
- Monitoring and logging setup
- Documentation and training

8.2 Release Timeline

Milestone	Date	Version
Alpha Release	Month 2	0.1
MVP Release	Month 3	1.0
Phase 2 Release	Month 6	1.5
Phase 3 Release	Month 9	2.0
Production Launch	Month 10	2.0

9. COST & RESOURCE ESTIMATION

9.1 Team Composition

Role	Count	Responsibility
Backend Developer	3	API development, database design
Frontend Developer	2	Web UI, state management
Mobile Developer	1	iOS/Android app (Phase 3+)

Role	Count	Responsibility
QA Engineer	1	Testing, automation
DevOps Engineer	1	Infrastructure, CI/CD, deployment
Project Manager	1	Planning, coordination
Product Manager	1	Requirements, prioritization
Designer	1	UI/UX design
Total	**11**	

Monthly Cost (Team): ~$100,000 – $150,000 (depending on location and seniority)

**9.2 Infrastructure Costs (Monthly)

Component	Cost	Notes
Cloud Hosting (AWS/GCP)	$2,000 – $5,000	EC2, RDS, S3
Database	$500 – $1,500	PostgreSQL managed service
Redis/Caching	$200 – $500	Elasticache
CDN	$100 – $300	CloudFront

Component	Cost	Notes
Message Queue	\$200 – \$500	RabbitMQ managed or SQS
Monitoring & Logs	\$300 – \$1,000	DataDog, ELK Stack
Email Service	\$50 – \$200	SendGrid
SMS/WhatsApp	\$500 – \$2,000	Twilio, MessageBird
Payment Gateway	2.9% + \$0.30 per transaction	Stripe
Domain & SSL	\$50 – \$100	Annual
Backup & DR	\$100 – \$300	Automated backups
Total	**\$4,000 – \$11,400/month**	

9.3 Software Licenses (Optional)

Software	Cost	Notes
Figma (Design)	\$12 – \$80/user/month	Design collaboration

Software	Cost	Notes
GitHub (Private Repo)	Free – $231/month	Code repository
Jira (Project Management)	Free – $1,458/month	Issue tracking, planning
Confluence (Documentation)	Free – $1,375/month	Internal documentation
Slack (Communication)	Free – $900+/month	Team communication

9.4 Total Estimated Cost (9 Months to Launch)

Category	Cost
Team Salary (9 months)	$900,000 – $1,350,000
Infrastructure (9 months)	$36,000 – $102,600
Third-Party Services	$30,000 – $50,000
Tools & Licenses	$5,000 – $10,000
Testing & QA	$10,000 – $15,000
Security & Compliance	$10,000 – $20,000
Contingency (10%)	$100,000 – $155,000

Category	Cost
TOTAL	$1,091,000 – $1,702,600

10. APPENDIX & GLOSSARY

10.1 Glossary

Term	Definition
Appointment	A scheduled consultation between a patient and a doctor
JWT	JSON Web Token; stateless authentication token
OAuth2	Open authorization protocol for secure third-party authentication
HIPAA	Health Insurance Portability and Accountability Act (US healthcare compliance)
GDPR	General Data Protection Regulation (EU data privacy law)
API	Application Programming Interface
PCI-DSS	Payment Card Industry Data Security Standard
SLA	Service Level Agreement; commitment to uptime/availability

Term	Definition
RPO	Recovery Point Objective; maximum acceptable data loss
RTO	Recovery Time Objective; maximum acceptable downtime
Microservices	Independent, loosely coupled services deployed separately
Monolith	Single, tightly coupled codebase deployed together
CI/CD	Continuous Integration/Continuous Deployment
Docker	Containerization platform for packaging applications
Kubernetes	Container orchestration platform for auto-scaling and management

10.2 Non-Functional Requirements Summary Table

Requirement	Specification
Uptime	99.5%
Page Load Time	≤ 3 seconds

Requirement	Specification
API Response	≤ 500ms (p95)
Concurrent Users	5,000 (MVP) → 50,000 (scaled)
Encryption	TLS 1.2+, AES-256
Authentication	JWT + OAuth2
Database	PostgreSQL, ACID compliance
Backup Frequency	Daily, 4-hour RPO
Code Coverage	≥80%
Accessibility	WCAG 2.1 AA

10.3 Key Integration Points

Mermaid

```
graph LR
    A["Web/Mobile App"] -->|REST API| B["API Gateway"]
    B --> C["Backend Services"]
    C --> D["PostgreSQL"]
    C --> E["Redis"]
    C --> F["RabbitMQ"]
    F --> G["Notification Worker"]
    G -->|SMS/WhatsApp| H["Twilio"]
    G -->|Email| I["SendGrid"]
```

```
    C -->|Payment| J["Stripe/Razorpay"]
    C -->|Files| K["AWS S3"]
```

10.4 Security Checklist

- [ ] All data encrypted in transit (HTTPS/TLS)
- [ ] All data encrypted at rest (AES-256)
- [ ] Passwords hashed with bcrypt (cost ≥12)
- [ ] SQL injection prevention (parameterized queries)
- [ ] XSS prevention (input validation, output encoding)
- [ ] CSRF protection (SameSite cookies, tokens)
- [ ] Rate limiting implemented
- [ ] API authentication via JWT
- [ ] Role-based access control (RBAC)
- [ ] Audit logging enabled
- [ ] Secrets managed securely (not in code)
- [ ] HIPAA/GDPR compliance verified
- [ ] Penetration testing completed
- [ ] Security headers configured (CSP, X-Frame-Options, etc.)

10.5 Testing Strategy

Level	Scope	Tools
Unit Testing	Individual functions/methods	Jest, JUnit
Integration Testing	API endpoints, database interactions	Supertest, Jest
E2E Testing	Complete user workflows	Cypress, Selenium
Performance Testing	Load, stress, endurance	JMeter, Locust
Security Testing	Vulnerability scanning, penetration testing	OWASP ZAP, Burp Suite

Target Coverage: ≥80% code coverage

10.6 Deployment Checklist

- [ ] Code review completed
- [ ] All tests passing (unit, integration, E2E)
- [ ] Security scan passed (SAST/DAST)
- [ ] Database migrations tested
- [ ] Backup verified
- [ ] Load balancer configured
- [ ] Monitoring/logging enabled

- [ ] SSL certificates valid
- [ ] Environment variables configured
- [ ] API documentation updated
- [ ] Runbooks prepared for common incidents
- [ ] Team trained on deployment process

10.7 Post-Launch Monitoring & Maintenance

Ongoing Tasks

1. **Weekly:** Review error logs, performance metrics
2. **Monthly:** Update dependencies, security patches
3. **Quarterly:** Review and update SLA, capacity planning
4. **Annually:** Penetration testing, HIPAA/GDPR audit

KPIs to Track

- Uptime percentage
- Average response time
- Error rate
- User satisfaction (NPS)
- Appointment booking conversion rate
- Payment success rate
- No-show rate
- Doctor and patient retention